6. Decide whether $x^2 \equiv 150 \pmod{1009}$ is solvable or not.

7. Find all primes p such that $x^2 \equiv 13 \pmod{p}$ has a solution.

***13.** Prove that there are infinitely many primes of each of the forms $3n + 1$ and $3n - 1$. (H)

14. Let p and q be *twin primes*, that is, primes satisfying $q = p + 2$. Prove that there is an integer a such that $p|(a^2 - q)$ if and only if there is an integer b such that $q|(b^2 - p)$. (There is a famous unsolved problem to prove that the number of pairs of twin primes is infinite. What *is* known is that the sum of the reciprocals of all twin primes is, if not a finite sum, certainly a convergent series; this result can be contrasted with Theorem 1.19. A proof of this result can be found in Chapter 15 of the book by Hans Rademacher, or in Chapter 6 of the 1977 book by W. J. LeVeque listed in the General References.)

***16.** Show that if $p = 2^{2^n} + 1$ is prime then 3 is a primitive root (mod p) and that 5 and 7 are primitive roots provided that $n > 1$.

4. Determine whether $x^4 \equiv 25 \pmod{1013}$ is solvable, given that 1013 is a prime.

9. For which positive integers n do there exist integers x and y with $(x, n) = 1$, $(y, n) = 1$, such that $x^2 + y^2 \equiv 0 \pmod{n}$?

***14.** Suppose that p is a prime, $p \equiv 1 \pmod 4$, and that $a^2 + b^2 = p$ with a odd and positive. Show that $\left(\frac{a}{p}\right) = 1$.

Theorem 3.7 *If Q is odd and $Q > 0$, then*

$$\left(\frac{-1}{Q}\right) = (-1)^{(Q-1)/2} \qquad \textit{and} \qquad \left(\frac{2}{Q}\right) = (-1)^{(Q^2-1)/8}$$

Theorem 3.8 *If P and Q are odd and positive and if $(P, Q) = 1$, then*

$$\left(\frac{P}{Q}\right)\left(\frac{Q}{P}\right) = (-1)^{\{(P-1)/2\}\{(Q-1)/2\}}.$$

2. Prove that the quadratic form $x^2 - 2xy + y^2$ has discriminant 0. Determine the class of integers represented by this form.

4. Use the binomial theorem to give a formula for positive integers x_k and y_k such that $(3 + 2\sqrt{2})^k = x_k + y_k\sqrt{2}$. Show that $(3 - 2\sqrt{2})^k = x_k - y_k\sqrt{2}$. Deduce that $x_k^2 - 2y_k^2 = 1$ for $k = 1, 2, 3, \cdots$. Show that $(x_k, y_k) = 1$ for each k. Show that $x_{k+1} = 3x_k + 4y_k$ and $y_{k+1} = 2x_k + 3y_k$ for $k = 1, 2, 3 \cdots$. Show that $\{x_k\}$ and $\{y_k\}$ are strictly increasing sequences. Conclude that the number 1 has infinitely many proper representations by the quadratic form $x^2 - 2y^2$.

1. Find a reduced form that is equivalent to the form $7x^2 + 25xy + 23y^2$.

3. Let x and y be integers. Show that there exist integers u and v such that $\begin{bmatrix} x & y \\ u & v \end{bmatrix} \in \Gamma$ if and only if $(x, y) = 1$.

4. Show that a binary quadratic form f properly represents an integer n if and only if there is a form equivalent to f in which the coefficient of x^2 is n. Use this and (3.3) to give a second proof of Theorem 3.13.

5. Show that $x^2 + 5y^2$ and $2x^2 + 2xy + 3y^2$ are the only reduced quadratic forms of discriminant -20. Show that the first of these forms does not represent 2, but that the second one does. Deduce that these forms are inequivalent, and hence that $H(-20) = 2$. Show that an odd prime p is represented by at least one of these forms if and only if $p \equiv 1, 3, 7$, or $9 \pmod{20}$.

6. Let $f(x, y) = ax^2 + bxy + cy^2$ and $g(x, y) = f(-x, y) = ax^2 - bxy + cy^2$. These forms represent precisely the same numbers, but they are not necessarily equivalent (because the determinant of the transformation has determinant -1). Show that $x^2 + xy + 2y^2$ is equivalent to $x^2 - xy + 2y^2$, but that $3x^2 + xy + 4y^2$ and $3x^2 - xy + 4y^2$ are not equivalent.

7. Let $f(x, y)$ be a quadratic form whose discriminant d is a positive perfect square. Show that f is equivalent to a form $ax^2 + bxy + cy^2$ for which $c = 0$ and $0 \leqslant a < |b|$. Deduce that there are only finitely many equivalence classes of forms of this discriminant. (H)